

TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Smart RISC-V 3D GPUs

Think Silicon has released a series of RISC-V 3D GPUs called the NEOX. The series includes NEOX G (graphics) and NEOX A (deep learning accelerator). These smart GPUs feature programmable compute shaders running on a real-time operating system and are supported by lightweight graphics and machine learning frameworks. The multi-threaded system can be customised for different applications—such as graphics, machine learning, vision/video processing, and general-purpose compute workloads—through configurable programming libraries using the same hardware blocks.

Think Silicon

<https://www.think-silicon.com>

ReRAM with dense memory

The MB85AS12MT from Fujitsu is a non-volatile IC capable of storing character data equivalent of 90 newspaper



pages. The 12Mbit ReRAM is said to have the largest memory density in ReRAM family. The IC comes in a compact package measuring approximately 2mm × 3mm and offers an outstandingly low level of reading current of 0.15mA. The size and current consumption make it suitable for use in wearable devices like smartwatches

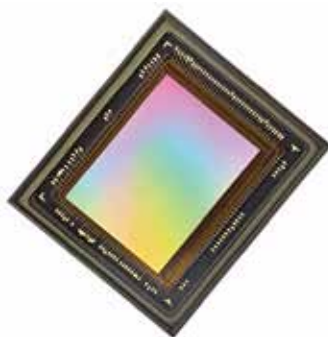
and wireless earphones.

Fujitsu Semiconductor Memory Solution

<https://www.fujitsu.com/jp/group/fsm/en>

Event based vision sensor

The Metavision sensor from Prophesee is an event based vision sensor designed for machine vision applications. The third-generation sensor is claimed to be the first of its kind in an



industry-standard package. Each pixel in the sensor can intelligently activate itself depending on the contrast change (movement) it detects, and each pixel embeds its own intelligence processing enabling it to activate itself independently. This makes the sensor highly power-efficient with just 3nW/event and 26mW at the sensor level. The sensor is capable of detecting images in areas with illumination level as low as 0.08lx. It can be used for ultra-high-speed counting, vibration measurement, kinematic monitoring, etc.

Prophesee

<https://www.prophesee.ai>

Multi-zone ToF ranging sensor

VL53L8 from STM is a second-generation, multi-zone, time-of-flight (ToF), laser-ranging sensor. It can measure absolute distance, irrespective of

colour and reflectance of the surface. It works at a fast speed and provides accurate ranging up to 400cm. The VL53L8 is also able to detect multiple



objects within the FoV with depth information for quicker touch-to-focus or mini depth map creation. It consumes 50% lower energy than its predecessor.

STMicroelectronics

https://www.st.com/content/st_com/en.html

Image sensor for mobile applications

The Topaz series of industrial CMOS sensors from Teledyne include 2MP (1920 × 1080) and 1.5MP (1920 × 800) resolution devices. The sensors are housed in a small 4.45mm wide chip-scale package. The Topaz sensors have an 8.5mm optical format, which is made possible due to the small 2.5µm global shutter pixel that employs in-

